

Injective PEPS Fundamental Theorem: Section 3

Route

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1 The source argument

The injective PEPS part of [MGRPG⁺18] proves the Fundamental Theorem for tensor networks on finite graphs without double edges and without self-loops. The proof is local. It does not slice a two-dimensional network into a long one-dimensional chain. Instead, it fixes an edge, blocks the graph around that edge into a three-site injective MPS, applies the one-dimensional injective theorem, and repeats this construction over all edges.¹

For a chosen edge $e = (u, v)$, all vertices other than u and v are grouped into the middle tensor. The two endpoint tensors and the middle tensor form a three-site MPS. Injectivity of the PEPS tensors implies injectivity of this blocked MPS. Since the two original PEPS represent the same state, the two blocked MPS represent the same state. The one-dimensional injective theorem therefore gives an algebra isomorphism between matrix insertions on the chosen bond. Equivalently, an arbitrary matrix X inserted on that bond in one network is matched by a matrix on the corresponding bond in the other network.

After choosing the resulting edge gauges for every edge and absorbing them into one tensor family, the paper obtains the following stronger comparison: for every edge and every matrix X , inserting X on that edge in the first PEPS gives the same state as inserting X on the same edge in the modified second PEPS.²

¹The edge-blocking construction is `Papers/1804.04964/paper_normal.tex:981--1009`. The equality of the two PEPS before blocking is at `Papers/1804.04964/paper_normal.tex:1010--1036`. The application of the three-site injective MPS theorem is at `Papers/1804.04964/paper_normal.tex:1037--1065`.

²This is the displayed equation labelled `eq:inj_equal_edge` in `Papers/1804.04964/paper_normal.tex:1037--1065`.

2 Current formal representation

The finite graph, edge, incident-edge, tensor, virtual configuration, state coefficient, equality of states, and vertex injectivity are represented in `TNLean/PEPS/Defs.lean`. Vertex injectivity is the linear independence of the local physical vectors indexed by local virtual configurations, matching the paper’s statement that each tensor is injective as a map from the virtual space to the physical space.

The local linear-algebraic realization of virtual operations at an injective vertex is represented in `TNLean/PEPS/VirtualInsertion.lean`. The chosen left inverse of the local tensor map and the resulting physical realization of local virtual endomorphisms correspond to the same observation used in the three-site MPS proof: a virtual operation can be represented by a physical operation on a neighboring site.

The edge-centered blocking data are represented in `TNLean/PEPS/Blocking.lean`. The existing declarations split local virtual configurations at a distinguished incident edge and split the vertex product into the left endpoint, the middle region, and the right endpoint. The edge-blocked coefficient is proved equal to the original PEPS coefficient.

The next formal layer is the matrix-insertion comparison. The declarations `TNLean.PEPS.EdgeInsertedBoundaryConfig`, `TNLean.PEPS.edgeBoundaryToInsertedBoundaryConfig`, `TNLean.PEPS.EdgeDiagonalInsertedBoundaryConfig`, `TNLean.PEPS.edgeBoundaryConfigEquivDiagonalInsertedBoundaryConfig`, `TNLean.PEPS.edgeMiddleConfigToOpenMiddleConfig`, `TNLean.PEPS.edgeOpenMiddleConfigToMiddleConfig`, `TNLean.PEPS.edgeMiddleConfigEquivOpenMiddleConfig`, `TNLean.PEPS.edgeOpenMiddleWeight`, and `TNLean.PEPS.edgeInsertedCoeff` represent the open edge contraction used when an arbitrary matrix is inserted on the distinguished edge. These definitions are the formal counterpart of the matrix insertion appearing before and after the equation labelled `eq:inj_equal_edge`. The two maps `TNLean.PEPS.edgeMiddleConfigToOpenMiddleConfig` and `TNLean.PEPS.edgeOpenMiddleConfigToMiddleConfig` have been packaged as the finite equivalence `TNLean.PEPS.edgeMiddleConfigEquivOpenMiddleConfig`. This isolates the reindexing needed for the identity-insertion specialization. The current Lean layer includes the corresponding middle-tensor reindexing theorem `TNLean.PEPS.edgeMiddleWeight_eq_edgeOpenMiddleWeight`. The diagonal summand has also been isolated in `TNLean.PEPS.edgeInsertedCoeff_identity_diagonal_summand`. The pointwise off-diagonal vanishing is recorded as `TNLean.PEPS.edgeInsertedCoeff_identity_offDiagonal_summand`;

what remains is to use `TNLean.PEPS.edgeBoundaryConfigEquivDiagonalInsertedBoundaryConfig` to reindex the finite inserted-boundary sum along the diagonal.

3 Remaining mathematical obligations

The following statements are still needed before the injective PEPS Fundamental Theorem can be proved in the manner of the paper.

1. The identity matrix coefficient must be used to collapse the two inserted bond indices to the diagonal. The diagonal summand theorem `thm:peps_edgeInsertedCoeff_identity_diagonal_summand` and the off-diagonal summand theorem `thm:peps_edgeInsertedCoeff_identity_offDiagonal_summand` reduce this to the diagonal equivalence `def:peps_edgeBoundaryConfigEquivDiagonalInsertedBoundaryConfig` and the finite diagonal reindexing theorem `thm:peps_edgeInsertedCoeff_identity`, identifying the matrix-insertion formalism with the ordinary three-block contraction when no nontrivial matrix is inserted. This remaining finite-sum step is tracked by issue #1372.
2. Vertex injectivity must be shown to pass to the edge-blocked three-site object. This is the formal version of the paper’s assertion that the regrouped MPS is injective. In the blueprint this is the not-yet-formalized theorem `thm:peps_edgeBlockedThreeSiteInjective`.
3. The two internal steps in Lemma `inj_isomorph` must be formalized for the edge-blocked three-site MPS. The virtual-to-physical realization $X \mapsto O_1, O_2$ is `thm:peps_virtualInsertionPhysicalRealization`; the converse recovery $O_1, O_2 \mapsto W$ is `thm:peps_physicalToVirtualInsertion`.
4. Equality of PEPS states must be upgraded to equality of edge-inserted coefficients after the three-site reduction. This is the step that produces the matrix-algebra isomorphism on the chosen bond. In the blueprint this is the not-yet-formalized theorem `thm:peps_edgeBlockedInsertionAlgebraIsomorphism`.
5. The algebra isomorphism must be converted into an invertible edge gauge, using the same finite-dimensional matrix-algebra argument as the one-dimensional injective theorem. In the blueprint this is the

not-yet-formalized theorem `thm:peps_edgeGaugeFromInsertionAlgebraIsomorphism`.

6. After all edge gauges have been absorbed into the second tensor family, producing the modified tensors $\tilde{B}_v$, the formalization must prove the analogue of `eq:inj_equal_edge`: arbitrary insertions of the same matrix on the same edge give equal PEPS states for the first tensor family and the modified second tensor family. In the blueprint the absorption step is the not-yet-formalized theorem `thm:peps_edgeGaugeAbsorption`, and the post-absorption insertion equality is the not-yet-formalized theorem `thm:peps_postAbsorptionEdgeInsertionEquality`.
7. The generalized two-injective-tensor comparison from `lem:inj_equal_tensors_2` must be formalized. It says that equality of all virtual-bond insertions for two pairs of injective tensors forces $A_1 = \lambda B_1$ and $A_2 = \lambda^{-1} B_2$. In the blueprint this is split into the scalar-reduction substep `thm:peps_twoInjectiveGaugeScalarReduction` and the not-yet-formalized theorem `thm:peps_twoInjectiveTensorInsertionComparison`.
8. The final two-tensor comparison must then be applied by blocking one vertex against its complement. The paper uses this to prove $A_i = \lambda_i \tilde{B}_i$ at each vertex, and then incorporates the scalars into the local gauges. In the blueprint this specialization is the separate not-yet-formalized theorem `thm:peps_oneVertexComplementComparison`.

4 Source-to-formalization ledger

The following ledger records the present status of the Section 3 proof route. It is meant to prevent the proof from drifting away from the source argument.

- `eq:block_to_mps` edge blocking: `edgeBlockedCoeff_eq_stateCoeff` is locally proved; the injectivity bridge is `thm:peps_edgeBlockedThreeSiteInjective`, tracked by issue #1366.
- Identity insertion on the chosen bond: `thm:peps_edgeInsertedCoeff_identity` is reduced to finite diagonal reindexing, tracked by issue #1372.
- Lemma `inj_isomorph`, $X \mapsto O_1, O_2$: `thm:peps_virtualInsertionPhysicalRealization`, tracked by issue #1369.

- Lemma `inj_isomorph`, $O_1, O_2 \mapsto W$: `thm:peps_physicalToVirtualInsertion`, tracked by issue #1370.
- Matrix-insertion algebra isomorphism: `thm:peps_edgeBlockedInsertionAlgebraIsomorphism`, tracked by issue #1367.
- Edge gauge from the algebra isomorphism: `thm:peps_edgeGaugeFromInsertionAlgebraIsomorphism`, tracked by issue #1368.
- Absorption of edge gauges into B : `thm:peps_edgeGaugeAbsorption`, tracked by issue #1363.
- Equation `eq:inj_equal_edge`: `thm:peps_postAbsorptionEdgeInsertionEquality`, tracked by issue #1364.
- Lemma `inj_equal_tensors_2`, scalar step: `thm:peps_twoInjectiveGaugeScalarReduction`, tracked by issue #1362.
- Lemma `inj_equal_tensors_2`, comparison: `thm:peps_twoInjectiveTensorInsertionComparison`, tracked by issue #1361.
- One vertex versus its complement: `thm:peps_oneVertexComplementComparison`, tracked by issue #1360.
- Final gauge equivalence and uniqueness: `thm:fundamentalTheorem_EPS_of_bondDim` and the uniqueness endpoint, tracked by issues #780 and #842.

5 Blueprint diagrams

The blueprint now uses semantic tensor-network commands for the Section 3 moves that should remain visually attached to the proof.

- `\TNPEPSEdgeBlockingReduction` depicts the chosen edge and the passage to the three-site chain. Its endpoint labels follow `eq:block_to_mps`: the two circled endpoints are A'_1 and A'_2 , while the boxed complement is A'_3 .
- `\TNPEPSEdgeInsertedCoeff` depicts the single edge-blocked contraction with one virtual matrix X inserted on the chosen bond. This is the diagram attached to the definition of the inserted-edge coefficient, before comparing two tensor networks.

- `\TNPEPSThreeSiteInsertionComparison` depicts the comparison of a virtual insertion in the two three-site chains, as in the statement of `lem:inj_isomorph`.
- `\TNPEPSInsertionPhysicalRealization` depicts the realization of a virtual insertion as a physical operation on either neighboring site.
- `\TNPEPSPhysicalToVirtualInsertion` depicts the converse step in which two neighboring physical operations determine a virtual operation on the shared bond.
- `\TNPEPSEdgeInsertionEquality` depicts the full PEPS comparison after the edge gauges have been absorbed into the second tensor family, matching the five-vertex graph and the distinguished $(2, 5)$ -edge insertion in `eq:inj_equal_edge`.
- `\TNPEPSEdgeGaugeAbsorption` depicts the local absorption of incident oriented edge gauges into a B -tensor, producing $\tilde{B}_v$.
- `\TNPEPSTwoInjectiveTensorInsertionComparison` depicts the generalized two-injective-tensor comparison with an arbitrary matrix inserted on a shared virtual bond.
- `\TNPEPSTwoInjectiveGaugeScalarReduction` depicts the residual virtual operators Z, U, W that appear after applying an inverse for one injective block in the proof of `lem:inj_equal_tensors_2`.
- `\TNPEPSOneVertexComplementComparison` depicts the final one-vertex-versus-complement comparison used to prove $A_v \propto \tilde{B}_v$. It follows the source paper's displayed comparison by drawing the one-vertex enlargement and the smaller injective region as tensor-network tensors, rather than as an auxiliary formal block.

These diagrams deliberately keep the local dot convention used in the blueprint, but they preserve the distinctions that matter in the paper: tensor sites, blocked regions, virtual matrix insertions, and physical operations are not drawn as the same object.

6 Conclusion

The current formalization has reached the edge-blocked coefficient and has begun the matrix-insertion layer needed for the Section 3 proof. It has not

yet proved the three-site injectivity transfer, the insertion comparison, the resulting edge gauge construction, or the final one-vertex-versus-complement comparison. Those are the remaining mathematical steps required to turn the current PEPS definitions into a proof of the injective PEPS Fundamental Theorem faithful to [MGRPG⁺18, Section 3].

References

- [MGRPG⁺18] András Molnár, José Garre-Rubio, David Pérez-García, Norbert Schuch, and J. Ignacio Cirac. Normal projected entangled pair states generating the same state. *New Journal of Physics*, 20(11):113017, 2018.